

StudyHub AWS Production & CI/CD Master Runbook

PRIMARY DOMAIN

https://studyhubonline.store

AWS REGION & ELASTIC IP

ap-south-1 (Mumbai) • 15.206.223.108

DEPLOYMENT STATUS

LIVE & SSL SECURED

DATABASE

AWS RDS PostgreSQL 18.3 (studyhub-prod-db)

1. Production Architecture Overview

```
[ Internet User / Browser ] | HTTPS (Port 443) ▼ [ AWS Elastic IP: 15.206.223.108 ] | [ Nginx Reverse Proxy + Let's Encrypt SSL ] | studyhubonline.store / www → 127.0.0.1:3000 (studyhub-web Next.js) | admin.studyhubonline.store → 127.0.0.1:3001 (studyhub-admin Next.js) | api.studyhubonline.store → 127.0.0.1:5000 (studyhub-api Express) | [ Internal Docker Network: studyhub-network ] | studyhub-redis (Port 6379) | studyhub-worker (Background Queues / BullMQ) | VPC Security Group Inbound Rule (Port 5432 with SSL) ▼ [ AWS RDS PostgreSQL: studyhub-prod-db:5432 ]
```

2. Cloud & Infrastructure Inventory

Resource	Identifier / Value	Purpose & Notes
AWS VPC	vpc-080ee16ec1b11057e	CIDR: 10.0.0.0/16
EC2 Server	i-0ccd30b73fe597ae7	Ubuntu 24.04 LTS (t3.medium , 30GB gp3)
Elastic IP	15.206.223.108	Static Public IPv4 attached to EC2
EC2 Security Group	sg-006dabca578c89482	Inbound: Port 22 (SSH), 80 (HTTP), 443 (HTTPS)
RDS PostgreSQL	studyhub-prod-db	Endpoint: studyhub-prod-db.cngoksag0z0e.ap-south-1.rds.amazonaws.com
RDS Security Group	sg-0ef77fb4a0998ff9a	Inbound: Port 5432 open to VPC CIDR 10.0.0.0/16
Swap Space	/swapfile (4GB)	Prevents OOM during Next.js production builds

3. DNS Configuration (GoDaddy)

Type	Host	Target Value	TTL	Status
A	@	15.206.223.108	1/2 Hour	ACTIVE
A	www	15.206.223.108	1/2 Hour	ACTIVE
A	admin	15.206.223.108	1/2 Hour	ACTIVE
A	api	15.206.223.108	1/2 Hour	ACTIVE

4. Local vs. Production Environment Comparison

Configuration	Local Development (<code>.env</code>)	Production EC2 (<code>.env.production</code>)
<code>NODE_ENV</code>	<code>development</code>	<code>production</code>
Web App URL	<code>http://localhost:3000</code>	<code>https://studyhubonline.store</code>
Admin Portal URL	<code>http://localhost:3001</code>	<code>https://admin.studyhubonline.store</code>
API Base URL	<code>http://localhost:5000/api/v1</code>	<code>https://api.studyhubonline.store/api/v1</code>
PostgreSQL Database	<code>localhost:5432/studyhub</code> (no SSL)	AWS RDS PostgreSQL (automatic SSL encryption)
Redis Cache	<code>redis://localhost:6379</code>	<code>redis://redis:6379</code> (Docker service name)

5. Database & Migration Commands Cheat Sheet

Action	Local Command (VS Code)	Production (EC2 / CI-CD)
Generate Prisma Client	<code>npm run db:generate</code>	Automatically run in Docker build
Create New Migration	<code>npm run db:migrate</code>	Created locally in VS Code
Deploy Migrations	<code>npm run db:deploy</code>	Automatically run by GitHub Actions CI/CD!
Database GUI (Studio)	<code>npm run db:studio</code>	N/A (local development only)
Seed Roles & Admin	<code>npm run seed:admin</code>	<code>docker compose -f docker-compose.prod.yml exec api npm run seed:admin</code>

6. Automated GitHub Actions CI/CD Pipeline

Pipeline Location: `.github/workflows/deploy-ec2.yml`

Required GitHub Secrets (One-Time Setup):

- `EC2_HOST` : `15.206.223.108`
- `EC2_USER` : `ubuntu`
- `EC2_SSH_KEY` : Full contents of `studyhub-key.pem`

Git Branching Workflow (No Direct Push to Main)

```
# 1. Start a feature in VS Code:  
git checkout -b feature/my-feature  
  
# 2. Commit and push:  
git add .  
git commit -m "feat: implement feature"  
git push -u origin feature/my-feature  
  
# 3. On GitHub: Open PR → Merge pull request  
# 4. GitHub Actions automatically connects to EC2, runs migrations, rebuilds & goes live!
```

7. Complete Troubleshooting Log & Fixes Applied

⚠️ 1. SSH Connection Timeout

Cause: EC2 was launched in a private subnet route table with no IGW route.

Solution: Associated subnet with `studyhub-public-rt` (Internet Gateway route).

⚠️ 2. Next.js Monorepo Standalone Output

Cause: Next.js builds failed in Docker without standalone directory.

Solution: Added `output: 'standalone'` in `apps/web/next.config.ts` and `apps/admin/next.config.ts`.

⚠️ 3. Memory Spikes & SSH Resets During Builds

Cause: Next.js compilations across 4 containers consumed 100% CPU/RAM.

Solution: Created a 4GB Swap file (`/swapfile`), added SSH keepalive, and built sequentially.

⚠️ 4. Admin Docker Build Missing public/ Folder

Cause: `apps/admin` lacked a public folder, breaking Docker `COPY`.

Solution: Added `RUN mkdir -p apps/admin/public` and created `.gitkeep`.

⚠️ 5. Prisma 7 Build-Time Generate Failure

Cause: Prisma 7 requires `DATABASE_URL` even during client generation.

Solution: Passed dummy `DATABASE_URL="postgresql://build:build@localhost:5432/build"`.

⚠️ 6. Node 24 ESM Module Resolution (API & Worker)

Cause: Node 24 strictly requires `.js` extensions on compiled relative imports.

Solution: Configured Dockerfiles to launch via `npm tsx --tsconfig ...`.

⚠️ 7. Missing package.json in Runtime Container

Cause: Omitted root `package.json`, breaking npm workspace resolution.

Solution: Added `COPY --from=build /app/package.json /app/package-lock.json ./`.

⚠️ 8. RDS PostgreSQL Port 5432 Inbound Rule

Cause: RDS Security Group blocked EC2 private IP (`10.0.87.55`).

Solution: Added Inbound Rule for PostgreSQL 5432 from VPC CIDR `10.0.0.0/16`.

⚠️ 9. Route Prefix Mismatch (/auth/login vs /api/v1/auth/login)

Cause: Frontend called `/auth/login` while backend mounted at `/api/v1/auth/login`.

Solution: Mounted routes at both `/api/v1` and `/`, and normalized client config URLs.

⚠️ 10. AWS RDS SSL Connection Rejection (no encryption)

Cause: AWS RDS strictly requires SSL; node-postgres attempted unencrypted connections.

Solution: Enabled automatic SSL (`ssl: { rejectUnauthorized: false }`) in `packages/database/src/client.ts`.

⚠️ 11. Missing User.avatarUrl Column

Cause: `avatarUrl` was in schema but had no migration generated.

Solution: Created migration `20260824000000_add_avatar_url` and added safe check to seed script.

⚠️ 12. Missing STUDENT Role & Admin 403 Forbidden Errors

Cause: Database was missing `STUDENT` role and `ROLE_PERMISSION_MANAGE` permissions for `ADMIN`.

Solution: Enhanced `seed-admin-user.ts` to bootstrap all 3 roles, 26 permissions, and grant full RBAC to Admin.

8. Verified Live Production Endpoints

- ✓ Main Web Application: <https://studyhubonline.store>
- ✓ Admin Management Portal: <https://admin.studyhubonline.store>
- ✓ Backend API Health: <https://api.studyhubonline.store/api/v1/health/ready>